



Operating system and system programming (25CS2104E)

Section : 12 Team : 14

LINUX FILE SYSTEM EXPLORER USING SYSTEM CALLS

Review-I

Under the Esteemed Guidance of

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Abstract

Linux File System Explorer Using System Calls is a Linux-based system programming project that provides a command-line interface for exploring and performing basic file-system operations using low-level Linux/POSIX system calls.

The project demonstrates file descriptors, file and directory management, permissions, metadata, and kernel-level resource access using C on Ubuntu/Linux. It uses APIs and system calls such as `open()`, `read()`, `write()`, `close()`, `lseek()`, `stat()`, `mkdir()`, `opendir()`, and `readdir()`.

Users can navigate directories, display file information, create files/directories, read and write files, and perform basic file-management operations.

Introduction

Problem Statement:

Many file-management applications hide the low-level operations performed by the operating system. This makes it difficult for students to understand how Linux handles files, directories, metadata, permissions, and file descriptors.

Proposed Solution:

Develop a simple command-line File System Explorer that directly uses Linux/POSIX system calls to perform common file and directory operations and demonstrate communication between a user program and the Linux kernel.

Objectives

Implement a Linux command-line file-system explorer using C and POSIX/Linux system calls.

Provide directory navigation, file creation, reading, writing, and listing.

Display file metadata such as size, type, permissions, and timestamps.

Demonstrate file descriptors, system calls, directory handling, and kernel interaction.

Hardware and Software Requirements

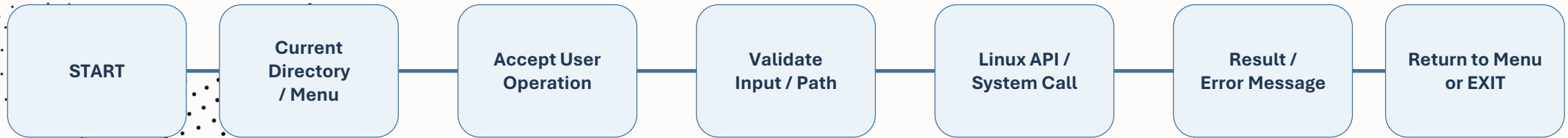
Requirements stated in the project problem statement

Tool / Platform / Software	Purpose
Linux / Ubuntu	Development and execution environment
C	Primary programming language for system-call implementation
GCC	Compile and build the C program
VS Code	Source-code editing and project development
Linux Terminal	Run, test, and demonstrate the application

Hardware: a development computer capable of running Ubuntu/Linux. Specific CPU/RAM/storage specifications are not stated in the problem statement.

System Architecture & Methodology

Application flow from user command to Linux/POSIX system call



Key Linux APIs / System Calls

read(), write() — read from and write to files

lseek() — move the file offset for random access

stat() — retrieve
open(), close() — open a file and release its file descriptor

size, type, permissions and timestamps

opendir(), readdir() — open and traverse directories

mkdir(), rmdir() — create and remove directories

Implementation Method

C program on Ubuntu/Linux

Menu-driven command-line interface

Map user commands to suitable Linux/POSIX APIs

Validate paths and handle errors using return values

Compile with GCC and test through the Linux terminal

Conclusion

The completed application is expected to provide a functional command-line File System Explorer through which users can list directory contents, navigate directories, create files and directories, read and write file contents, inspect file metadata, and perform selected basic file-management operations.

The project provides practical experience with Linux system calls, POSIX APIs, file descriptors, directory structures, permissions, file metadata, and connecting C programs with Linux kernel services.

GitHub Repository: [KLH_CSIT_2026-27_14_FileSystemExplorer](#)

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THANK YOU

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